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Truncated tetrahedron

In geometry, the **truncated tetrahedron** is an Archimedean solid. It has 4 regular hexagonal faces, 4 equilateral triangle faces, 12 vertices and 18 edges (of two types). It can be constructed by truncating all 4 vertices of a regular tetrahedron at one third of the original edge length.

A deeper truncation, removing a tetrahedron of half the original edge length from each vertex, is called rectification. The rectification of a tetrahedron produces an octahedron.[1]

A *truncated tetrahedron* is the Goldberg polyhedron $G_{III}(1,1)$, containing triangular and hexagonal faces.

A *truncated tetrahedron* can be called a **cantic cube**, with Coxeter diagram, , having half of the vertices of the cantellated cube (rhombicuboctahedron), . There are two dual positions of this construction, and combining them creates the uniform compound of two truncated tetrahedra.

Area and volume

The area A and the volume V of a truncated tetrahedron of edge length a are:

$$\begin{aligned} A &= 7\sqrt{3}a^2 && \approx 12.124\,355\,65a^2 \\ V &= \tfrac{23}{12}\sqrt{2}a^3 && \approx 2.710\,575\,995a^3. \end{aligned}$$

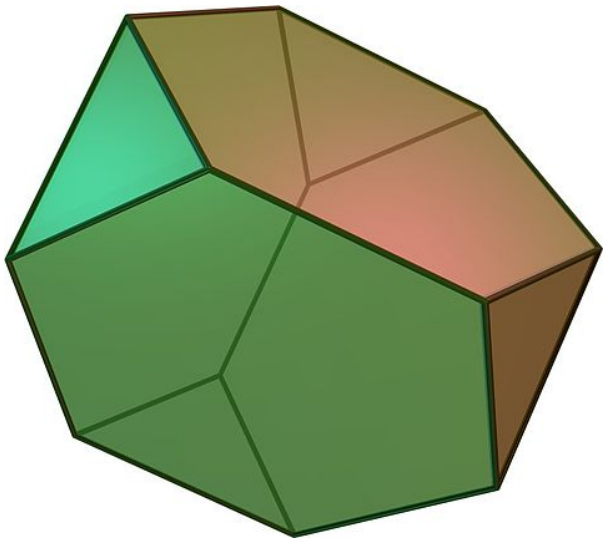
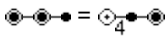
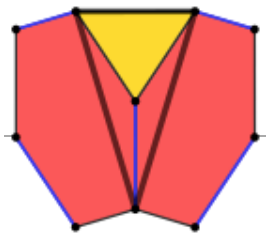
Densest packing

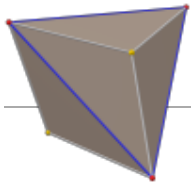
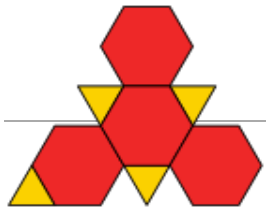
The densest packing of the Archimedean truncated tetrahedron is believed to be $\Phi = \frac{207}{208}$, as reported by two independent groups using Monte Carlo methods.[2][3] Although no mathematical proof exists that this is the best possible packing for the truncated tetrahedron, the high proximity to the unity and independency of the findings make it unlikely that an even denser packing is to be found. In fact, if the truncation of the corners is slightly smaller than that of an Archimedean truncated tetrahedron, this new shape can be used to completely fill space.[2]

Cartesian coordinates

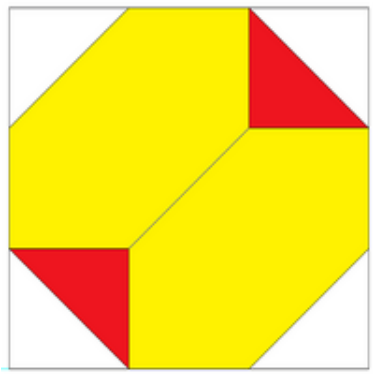
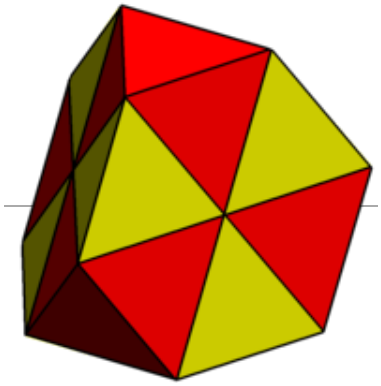
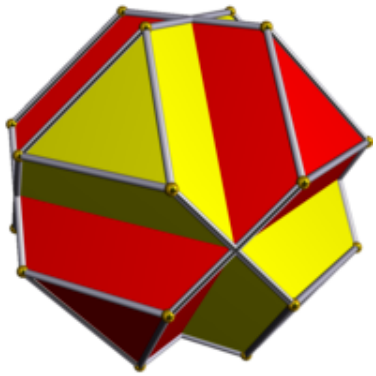
Cartesian coordinates for the 12 vertices of a truncated tetrahedron centered at the origin, with edge length √8, are all permutations of (±1,±1,±3) with an even number of minus signs:

- (+3,+1,+1), (+1,+3,+1), (+1,+1,+3)
- (-3,-1,+1), (-1,-3,+1), (-1,-1,+3)
- (-3,+1,-1), (-1,+3,-1), (-1,+1,-3)
- (+3,-1,-1), (+1,-3,-1), (+1,-1,-3)

Truncated tetrahedron	
 (Click here for rotating model)	
Type	Archimedean solid Uniform polyhedron
Elements	F = 8, E = 18, V = 12 (χ = 2)
Faces by sides	4{3}+4{6}
Conway notation	tT
Schläfli symbols	t{3,3} = h_2\{4,3\}
	$t_{0,1}$\{3,3\}
Wythoff symbol	2 3 | 3
Coxeter diagram	
Symmetry group	T_d, A_3, [3,3], (*332), order 24
Rotation group	T, $[3,3]^+$, (332), order 12
Dihedral angle	3-6: 109°28′16″ 6-6: 70°31′44″
References	U_{02}, C_{16}, W_6
Properties	Semiregular convex
Colored faces	 3.6.6 (Vertex figure)

 Triakis tetrahedron (dual polyhedron)	 Net

3D model of a truncated tetrahedron

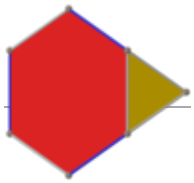
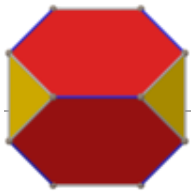
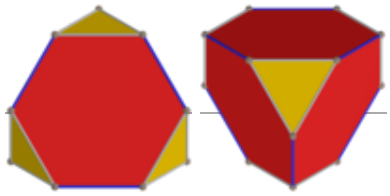
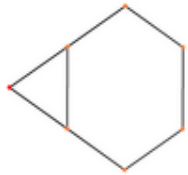
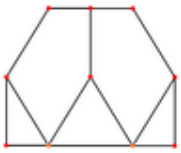
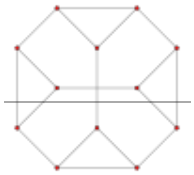
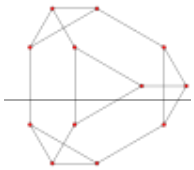
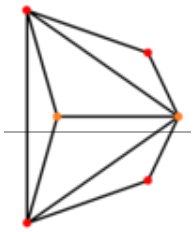
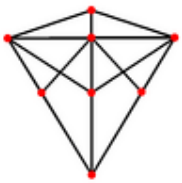
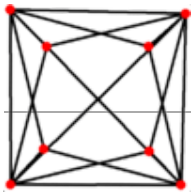
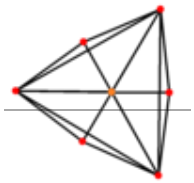
		
Orthogonal projection showing Cartesian coordinates inside its bounding box: (±3,±3,±3).	The hexagonal faces of the truncated tetrahedra can be divided into six coplanar equilateral triangles. The four new vertices have Cartesian coordinates: (-1,-1,-1), (-1,+1,+1), (+1,-1,+1), (+1,+1,-1). As a solid, this can represent a 3D dissection, making four red octahedra and six yellow tetrahedra.	The set of vertex permutations (±1,±1,±3) with an odd number of minus signs forms a complementary truncated tetrahedron, and combined they form a uniform compound polyhedron.

Another simple construction exists in 4-space as cells of the truncated 16-cell, with vertices as coordinate permutation of:

(0,0,1,2)

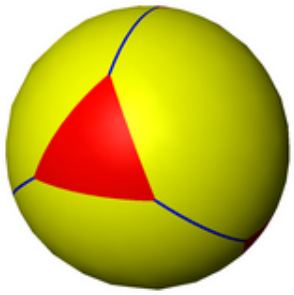
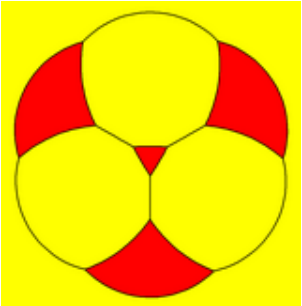
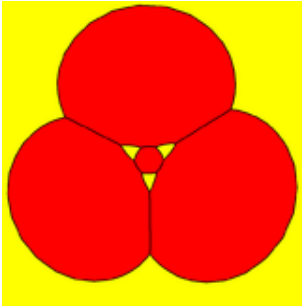
Orthogonal projection

Orthogonal projection

Centered by	Edge normal	Face normal	Edge	Face
Wireframe				
Wireframe				
Dual				
Projective symmetry	[1]	[1]	[4]	[3]

Spherical tiling

The truncated tetrahedron can also be represented as a spherical tiling, and projected onto the plane via a stereographic projection. This projection is conformal, preserving angles but not areas or lengths. Straight lines on the sphere are projected as circular arcs on the plane.

	 triangle-centered	 hexagon-centered
Orthographic projection	**Stereographic projections**	

Friauf polyhedron

A lower symmetry version of the truncated tetrahedron (a truncated tetragonal disphenoid with order 8 D_{2d} symmetry) is called a **Friauf polyhedron** in crystals such as complex metallic alloys. This form fits 5 Friauf polyhedra around an axis, giving a 72-degree dihedral angle on a subset of 6-6 edges. It is named after J. B. Friauf and his 1927 paper "The crystal structure of the intermetallic compound $MgCu_2$".[4]

Uses

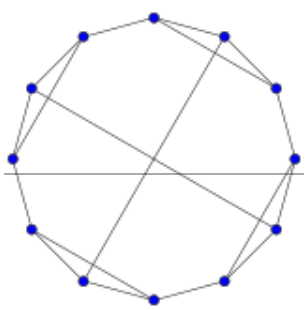
Giant truncated tetrahedra were used for the "Man the Explorer" and "Man the Producer" theme pavilions in Expo 67. They were made of massive girders of steel bolted together in a geometric lattice. The truncated tetrahedra were interconnected with lattice steel platforms. All of these buildings were demolished after the end of Expo 67, as they had not been built to withstand the severity of the Montreal weather over the years. Their only remnants are in the Montreal city archives, the Public Archives Of Canada and the photo collections of tourists of the times.[5]

The Tetraminx puzzle has a truncated tetrahedral shape. This puzzle shows a dissection of a truncated tetrahedron into 4 octahedra and 6 tetrahedra. It contains 4 central planes of rotations.

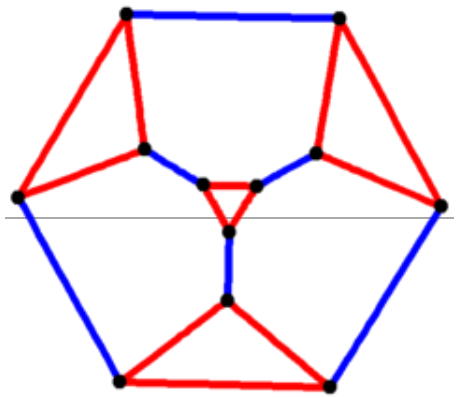
Truncated tetrahedral graph

In the mathematical field of graph theory, a **truncated tetrahedral graph** is an Archimedean graph, the graph of vertices and edges of the truncated tetrahedron, one of the Archimedean solids. It has 12 vertices and 18 edges.[7] It is a connected cubic graph,[8] and connected cubic transitive graph.[9]

Circular

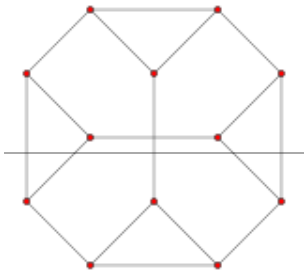
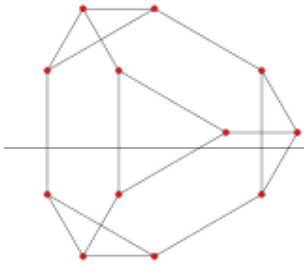


Truncated tetrahedral graph



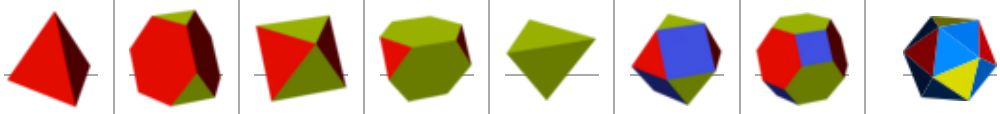
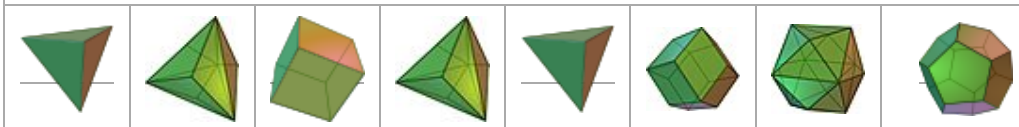
3-fold symmetry

Vertices	12[6]
Edges	18
Radius	3
Diameter	3[6]
Girth	3[6]
Automorphisms	24 (S_4)[6]
Chromatic number	3[6]
Chromatic index	3[6]

Orthographic projections	
	
4-fold symmetry	3-fold symmetry

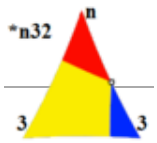
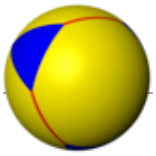
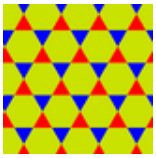
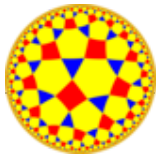
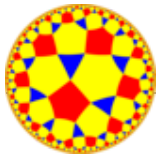
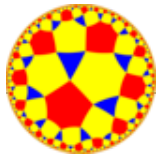
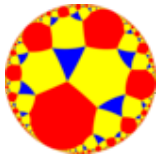
Properties	Hamiltonian, regular, 3-vertex-connected, planar graph
Table of graphs and parameters	

Related polyhedra and tilings

Family of uniform tetrahedral polyhedra							[show]
Symmetry: [3,3], (*332)							$[3,3]^+$, (332)
							
{3,3}	t{3,3}	r{3,3}	t{3,3}	{3,3}	rr{3,3}	tr{3,3}	sr{3,3}
Duals to uniform polyhedra							
							
V3.3.3	V3.6.6	V3.3.3.3	V3.6.6	V3.3.3	V3.4.3.4	V4.6.6	V3.3.3.3.3

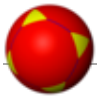
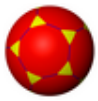
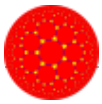
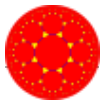
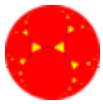
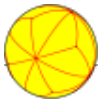
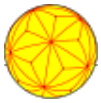
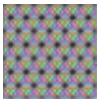
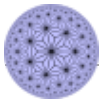
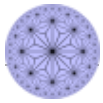
It is also a part of a sequence of cantic polyhedra and tilings with vertex configuration 3.6.*n*.6. In this wythoff construction the edges between the hexagons represent degenerate digons.

**n*33 orbifold symmetries of cantic tilings: *3.6.n.6*

	Orbifold *n32	Spherical	Euclidean	Hyperbolic			Paracompact
		*332	*333	*433	*533	*633...	*∞33
Cantic figure							
Vertex		3.6.**2**.6	3.6.**3**.6	3.6.**4**.6	3.6.**5**.6	3.6.**6**.6	3.6.∞.6

Symmetry mutations

This polyhedron is topologically related as a part of sequence of uniform truncated polyhedra with vertex configurations (3.2*n*.2*n*), and [*n*,3] Coxeter group symmetry.

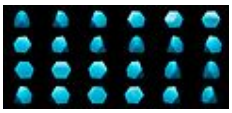
*n32 symmetry mutation of truncated spherical tilings: t{n,3}								[show]
Symmetry *n32 [n,3]	Spherical				Euclid.	Compact hyperb.		Paraco.
	*232 [2,3]	*332 [3,3]	*432 [4,3]	*532 [5,3]	*632 [6,3]	*732 [7,3]	*832 [8,3]...	*∞32 [∞,3]
Truncated figures								
Symbol	t{2,3}	t{3,3}	t{4,3}	t{5,3}	t{6,3}	t{7,3}	t{8,3}	t{∞,3}
Triakis figures								
Config.	V3.4.4	V3.6.6	V3.8.8	V3.10.10	V3.12.12	V3.14.14	V3.16.16	V3.∞.∞

Examples

drawing in De divina proportione (1509)

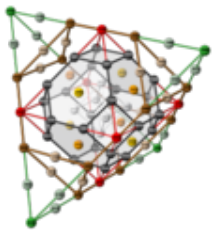
drawing in Perspectiva Corporum Regularium (1568)

crystal model



photos from different perspectives (Matemateca)

4-sided die



12 permutations of $(4, 2, 0, 0)$ (brown)

See also

- Quarter cubic honeycomb – Fills space using truncated tetrahedra and smaller tetrahedra
- Truncated 5-cell – Similar uniform polytope in 4-dimensions
- Truncated triakis tetrahedron
- Triakis truncated tetrahedron

- Octahedron – a rectified tetrahedron

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5. "Expo 67 - Man the Producer - page 1" (http://expo67.ncf.ca/man_the_producer_p1.html).
6. An Atlas of Graphs, page=172, C105
7. An Atlas of Graphs, page 267, truncated tetrahedral graph
8. An Atlas of Graphs, page 130, connected cubic graphs, 12 vertices, C105
9. An Atlas of Graphs, page 161, connected cubic transitive graphs, 12 vertices, Ct11

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- Read, R. C.; Wilson, R. J. (1998), *An Atlas of Graphs*, Oxford University Press

External links

- Weisstein, Eric W., "Truncated tetrahedron (https://mathworld.wolfram.com/TruncatedTetrahedron.html)" ("Archimedean solid (http://mathworld.wolfram.com/ArchimedeanSolid.html)") at *MathWorld*.
  - Weisstein, Eric W. "Truncated tetrahedral graph" (https://mathworld.wolfram.com/TruncatedTetrahedralGraph.html). *MathWorld*.
- Klitzing, Richard. "3D convex uniform polyhedra x3x3o - tut" (https://bendwavy.org/klitzing/dimensions/polyhedra.htm).
- Editable printable net of a truncated tetrahedron with interactive 3D view (http://www.dr-mikes-math-games-for-kids.com/polyhedral-nets.html?net=3cqTmfu7gdEZ8I7kRUVvji6qxBATVSp2WpmIWGx7l7pWe7bveylFxv3piHnPNZN&name=Truncated+Tetrahedron#applet)
- The Uniform Polyhedra (http://www.mathconsult.ch/showroom/unipoly/)
- Virtual Reality Polyhedra (http://www.georgehart.com/virtual-polyhedra/vp.html) The Encyclopedia of Polyhedra

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